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G H Raisoni International Skill Tech University, Pune

S.Y. B.Tech (All Branch)

(TERM-III)

CAE I- WINTER 2025 (2024 Pattern)

Discrete Mathematics and Graph Theory (25BTBSL301)

(Time: - 01 Hour) (Max. Marks: 20)

Instructions to the students:

i) All questions are compulsory. ii) Neat diagrams must be drawn wherever necessary. iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed v) Assume suitable data, if necessary.

Q.No	Question	Marks	CO	BL
Q1.(a)	i) Define partition and covering of a set with examples. ii) Let A= {1,2,3} and relation R={(1,1),(1,2),(2,3),(3,3)} find the relation matrix and draw a digraph and check whether is it reflexive..	[5]	1	1
b)	Find relation R of the set ({1,2,3,4,5,7,11,12,13,24,48,72}, |). And then draw Hasse diagram of R. { Here | denoted Divisibility means $a\|b \Rightarrow b = ak$ *or a divides b.*}	[5]	1	2
	OR			
Q 2.(a)	i)Using Mathematical Induction prove that for all natural numbers $n \geq 1$, $1 + 2 + 3 + \ldots + n = \frac{n(n+1)}{2}$ ii)Let U = {a, b, c, d, e, f, g, h}and set A = {a, b, c, d, f, g} and B = {a, b, e, g, h} Find AUB, $A \cap B'$, AΔB and Represent in 0's and 1's form.	[5]	1	2
(b)	Are the following relations equivalent or not? 1) Set of parallel lines 2) Set of perpendicular lines	[5]	1	1
Q 3.(a)	1) State Lagrange's theorem. 2) Define left and right coset. And find left cosets of Z_8 with respective subgroup generated by $< 2 >$ *and* $< 8 >$ *or* $H = 2Z$ *and* $H = 8Z$	[5]	2	3
(b)	Show that $G = \{1, -1, i, -i\}$ is Abelian group under multiplication. Also find the order of each element.	[5]	2	3
	OR			
Q. 4(a)	The subgroup $H = \{(1), (1\,2\,3), (1\,3\,2)\}$ of S_3. Show that H is Normal Subgroup	[5]	2	2
(b)	Find $\alpha\beta$ and $\alpha^{-1}\alpha\beta^{-1}$ where $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 3 & 5 & 1 \end{pmatrix}$ and $\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 1 & 2 & 3 \end{pmatrix}$.	[5]	2	3

*****End of Paper *****